Sakib Hassan

Education

University at Buffalo — **Jan. 2022 – May 2026**
Bachelor of Science in Computer Science with a Minor in Mathematics — *Buffalo, NY*

The Chinese University of Hong Kong (CUHK) — **Aug. 2025 – Dec. 2025**
Study Abroad Program — *Hong Kong*

Technical Skills

Languages (by familiarity): Python (Advanced), Java (Advanced), SQL (Proficient), C (Proficient), JavaScript (Intermediate), HTML/CSS (Intermediate)
Developer Tools: VS Code, Git, Google Colab, Anaconda, Docker, Unity, IntelliJ, Emacs, Xpra
Technologies/Frameworks: PyTorch, CUDA, TensorFlow, Postgres, Streamlit, GitHub

Work Experience

University at Buffalo — **January 2023 – May 2023**
Software Engineer Intern — *Buffalo, NY*

- Developed a React and WebSocket-based app for real-time 3D sensor data visualization with dynamic user interactions.
- Integrated secure authentication with Amazon Cognito for enhanced security.
- Designed a DynamoDB solution with Python for efficient data transformation and epoch-based queries.
- Built a high-performance Python WebSocket server on EC2 for seamless IoT communication.

Projects

Denoised Reinforcement Learning for Optimal Trade Execution | *Python, PyTorch, RL* — **Sep. 2025 – Dec. 2025**

- Developed AI-driven trade execution strategies using diffusion models and reinforcement learning on minute-level QQQ market data.
- Implemented a Diffusion-QL policy and compared it with PPO agents trained on historical and synthetic trajectories.
- Built a market simulation framework incorporating market impact, liquidity constraints, and participation caps based on the Almgren–Chriss model.
- Evaluated performance across 486 trading windows, achieving consistent improvements over TWAP-style baseline execution.

Multi-Agent Reinforcement Learning in Adversarial Environment | *Python, PyTorch, RL* — **November 2024**

- Implemented multi-agent Deep Q-Network (DQN) policies in a competitive environment using the PettingZoo Simple Adversary simulation.
- Designed shared policy and replay buffer mechanisms for cooperative agents while training an independent adversarial agent.
- Modeled adversarial and cooperative reward structures to study strategic behavior in partially observable multi-agent settings.
- Analyzed learning stability and agent performance across 2,000+ training episodes using reward trend evaluation.

Semantic Search Chatbot | *Python, Streamlit* — **September 2023**

- Built an AI-powered text analysis tool handling PDFs and raw text using LangChain and OpenAI embeddings for semantic retrieval.
- Enabled natural language queries for efficient information extraction across unstructured documents.
- Implemented document chunking, embedding pipelines, and vector similarity search to support scalable retrieval across large text corpora.
- Evaluated retrieval quality through query testing and prompt tuning to improve answer relevance and system reliability.

Awards / Extracurricular

UB Hacking 2022 Winner — **November 2022**
Financial Game — *University at Buffalo*

- Won Best Freshman Hack and M&T Tech Awards.
- Built an educational financial literacy game using Unity.
- Developed a supporting website with HTML and CSS.